THE LANGUAGE OF

MATHEMATICS

General notation

$=$	Equal to
$\neq$	Not equal to
$\approx$	Approximately equal to
$<,\ >$	Less than; greater than
$\leq,\ \geq$	Less than or equal to; greater than or equal to
$\pm$	Plus or minus
$\lvert a \rvert$	Absolute value
$\sqrt{a}$	Square root
$\sum$	Summation symbol
$n!$	Factorial notation

Sets and number systems

$\{a, b, c\}$	Set with three elements
$\varnothing$	Empty set
$\in,\ \notin$	Is an element of; is not an element of
$\subseteq$	Is a subset of
$\cup,\ \cap$	Union; intersection
$\mathbb{N},\ \mathbb{Z}$	Natural numbers; integers
$\mathbb{Q},\ \mathbb{R}$	Rational numbers; real numbers
$(a, b),\ [a, b]$	Open interval; closed interval

Logic

$\Rightarrow$	Implies
$\Leftrightarrow$	If and only if
$\forall$	For every
$\exists$	There exists
$\neg$	Logical negation

Probability and events

Ω	Sample space: all possible outcomes
ω	An individual outcome
$A = \{1, 3, 5\}$	Event shown as a set of outcomes
$A \subseteq \Omega$	An event is a subset of the sample space
$P(A)$	Probability of event A
A^c	Complement of event A
$A \cup B$	Event A or B (or both)
$A \cap B$	Both events A and B
$P(A \mid B)$	Probability of A given B

Function notation

f	Name of a function
$f(x)$	Value of f at input x
$f : A \to B$	Function from set A to set B
$x \mapsto f(x)$	Input-to-output mapping
$\operatorname{dom}(f)$	Domain of a function

Letters and Greek symbols

$a, b, c, \ldots$	Common lowercase Latin letters
$A, B, C, \ldots$	Common uppercase Latin letters
x, y, z	Other common Latin letters
$\alpha,\ \beta,\ \gamma$	Alpha; beta; gamma
$\delta,\ \Delta$	Delta; uppercase delta
$\theta,\ \Theta$	Theta; uppercase theta
$\lambda,\ \Lambda$	Lambda; uppercase lambda
$\mu,\ \sigma,\ \Sigma$	Mu; sigma; uppercase sigma
$\pi,\ \Pi$	Pi; uppercase pi
$\phi,\ \Phi$	Phi; uppercase phi
$\omega,\ \Omega$	Omega; uppercase omega

Symbols acquire meaning from their mathematical context.